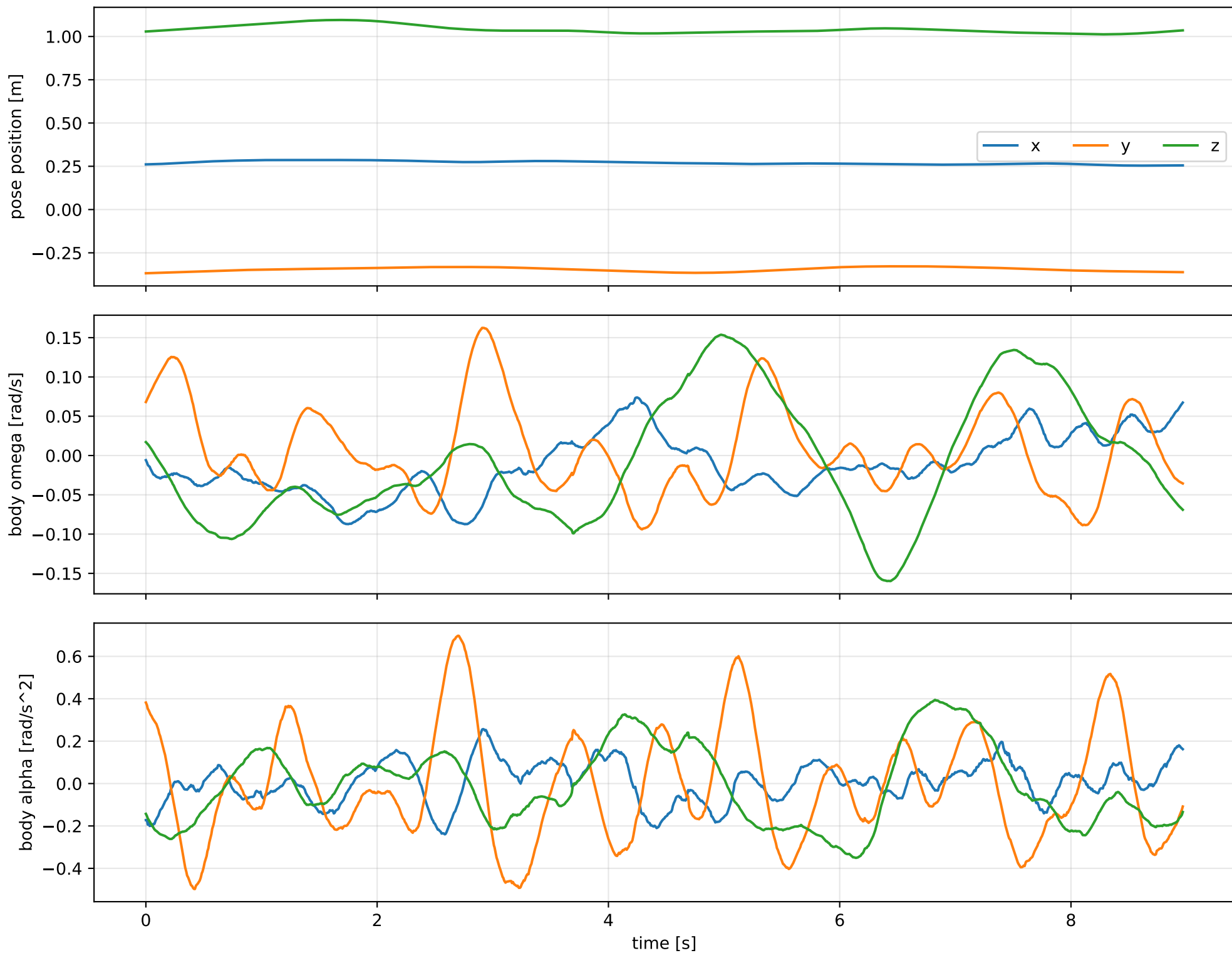
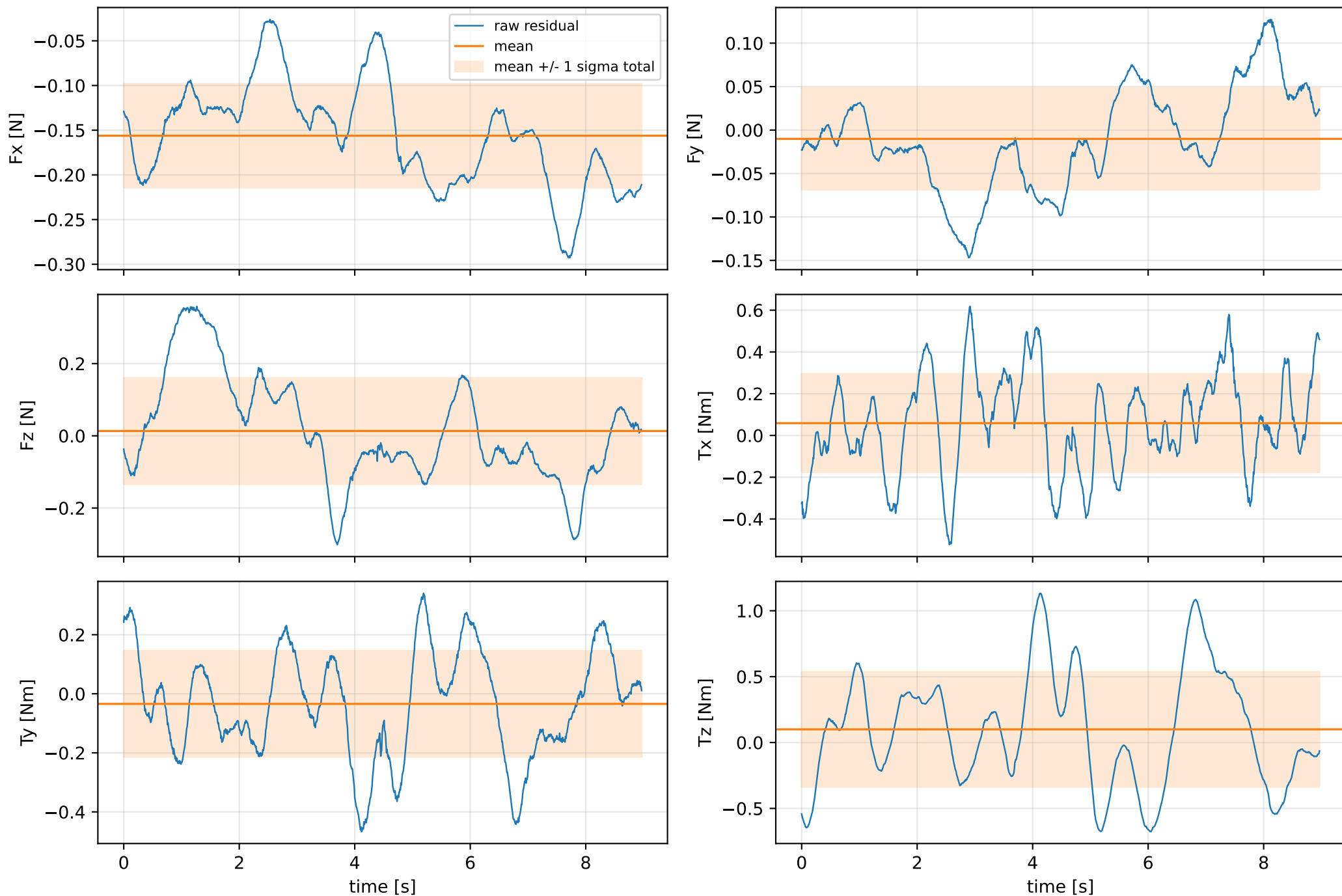


cog_all_nominal: SG trajectory and derived motion



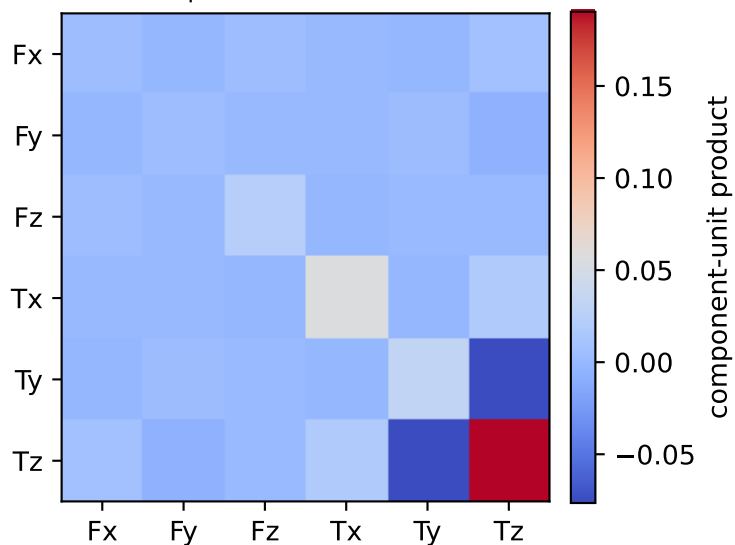
cog_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



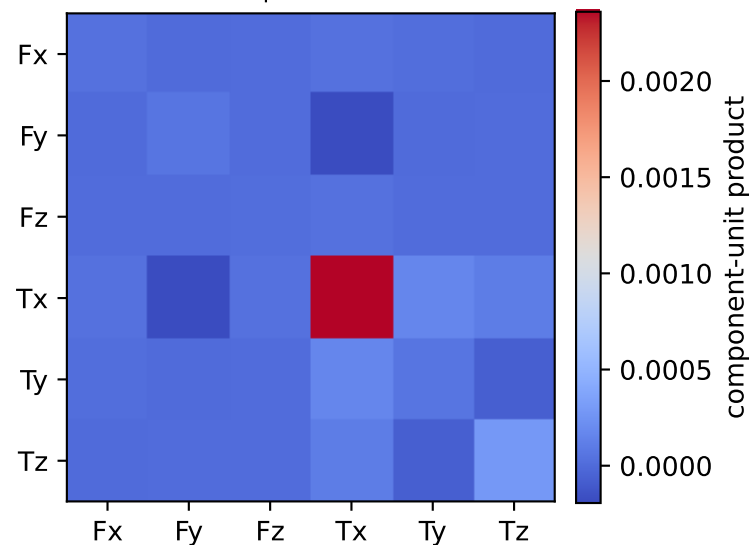
vector RMS about zero: force 0.23035 N, torque 0.54073 Nm; component std about mean: force [0.0581 0.0584 0.1471], torque [0.2355 0.1797 0.4362]

cog_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.005 0.022 0.052 0.223]

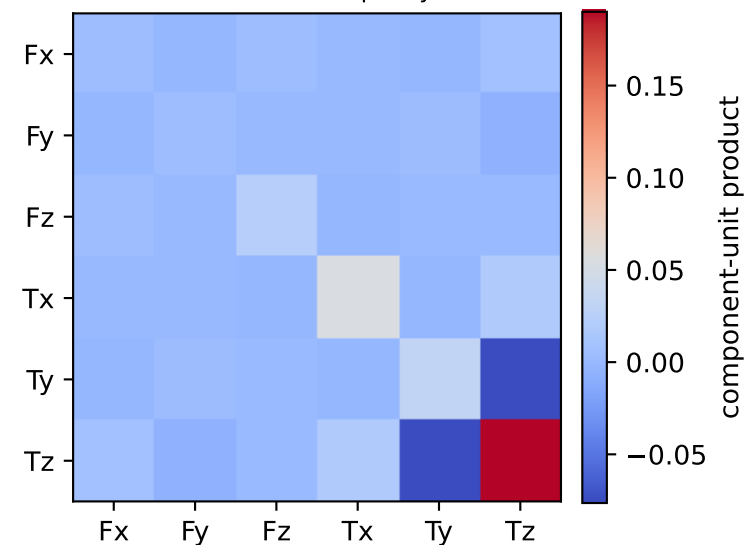
empirical total covariance



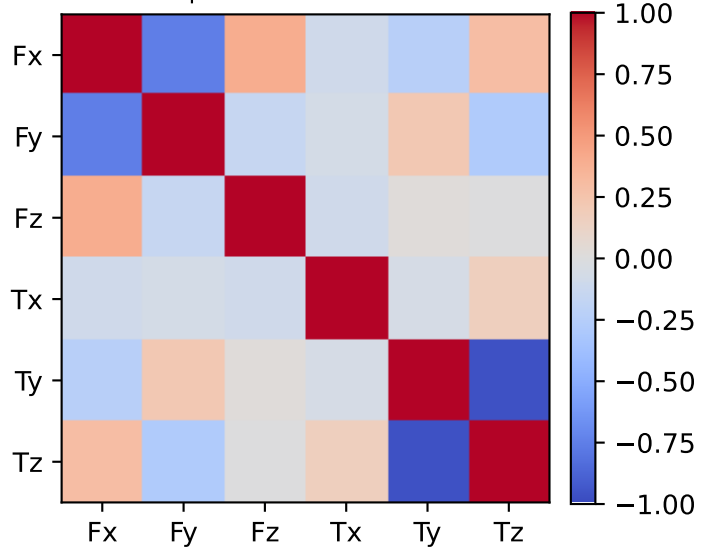
mean full-SG prediction covariance



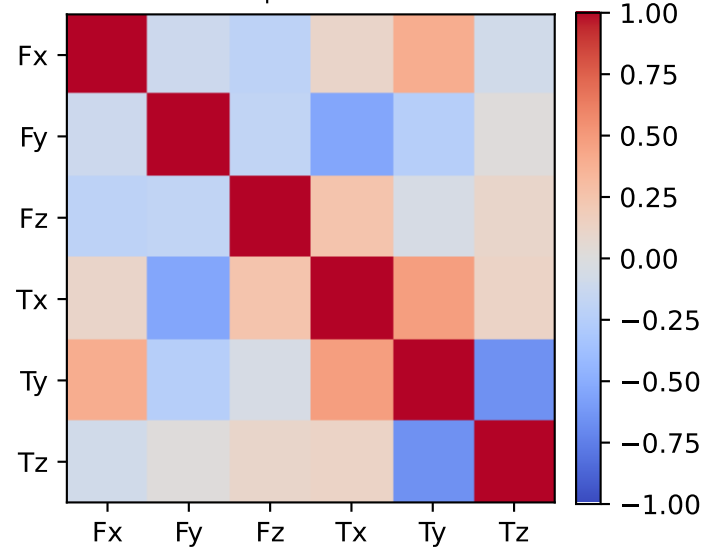
PSD excess model discrepancy covariance



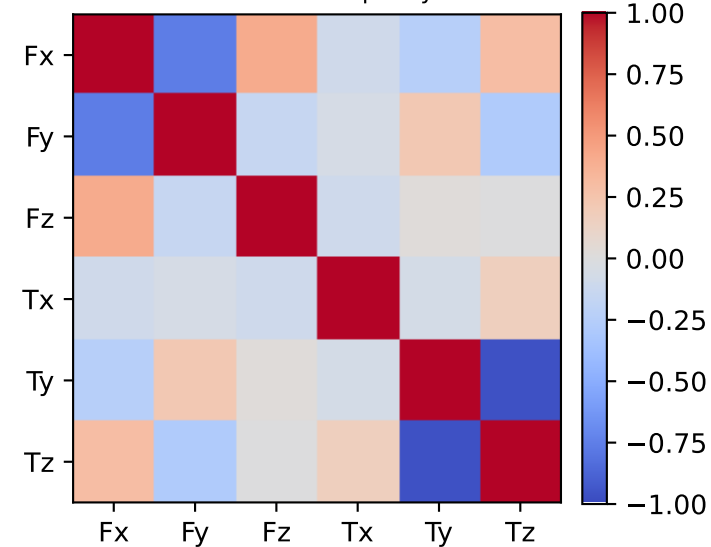
empirical total correlation



mean full-SG prediction correlation

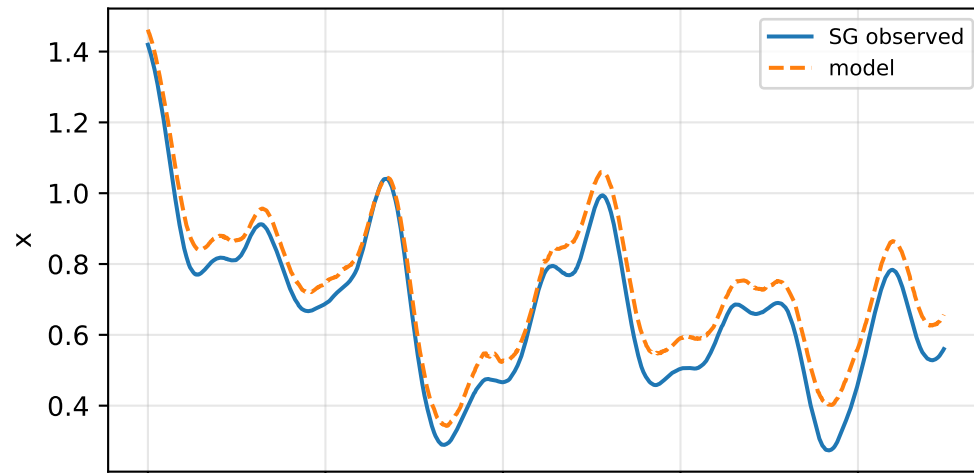


PSD excess model discrepancy correlation

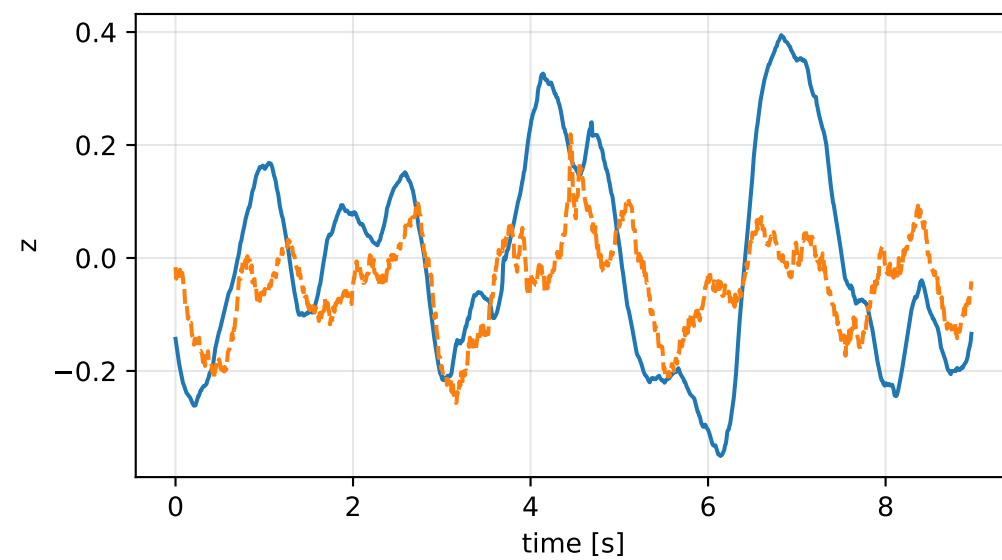
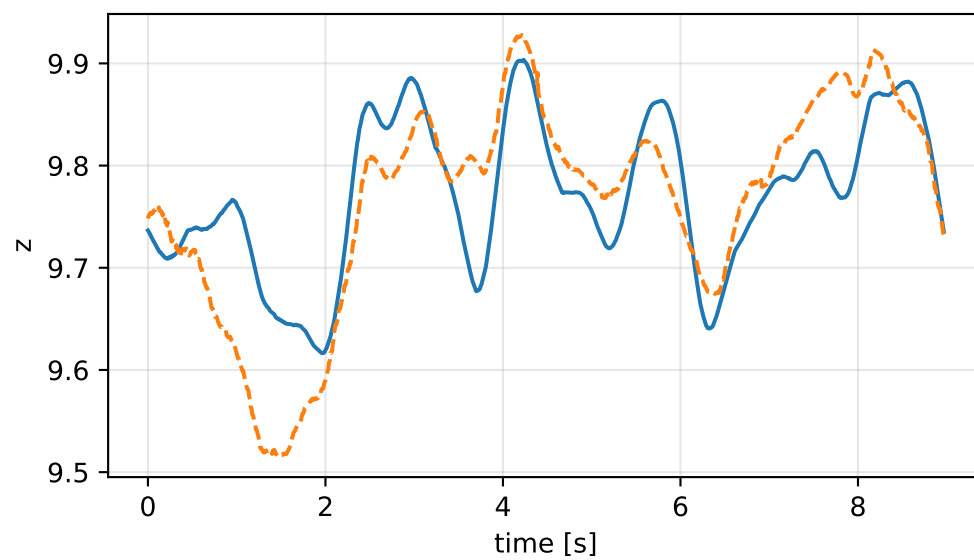
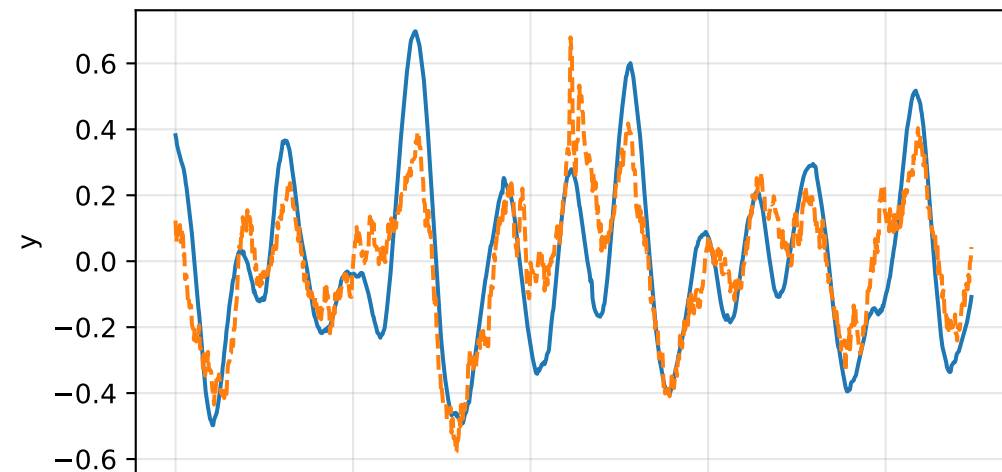
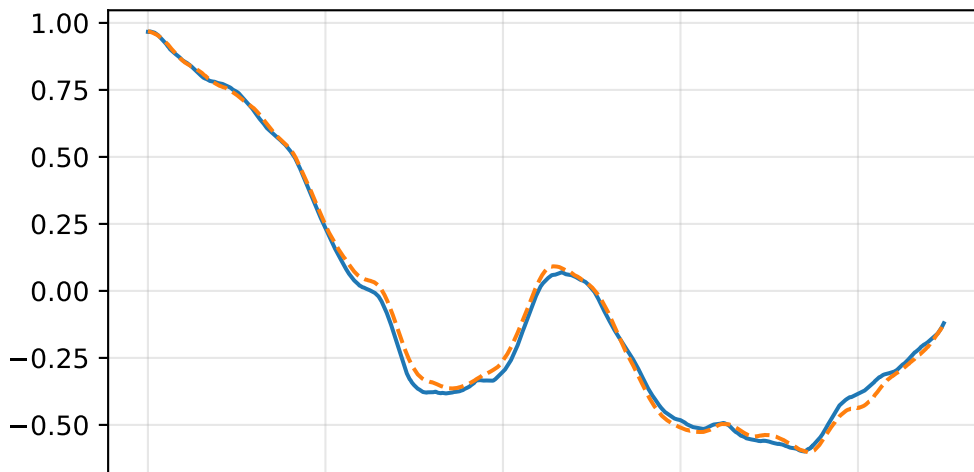
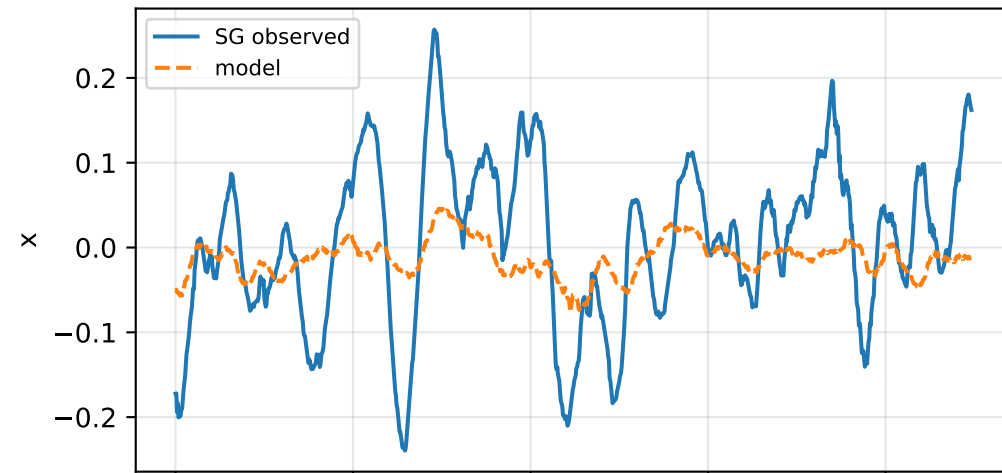


cog_all_nominal: acceleration residual objective

specific acceleration [m/s²]

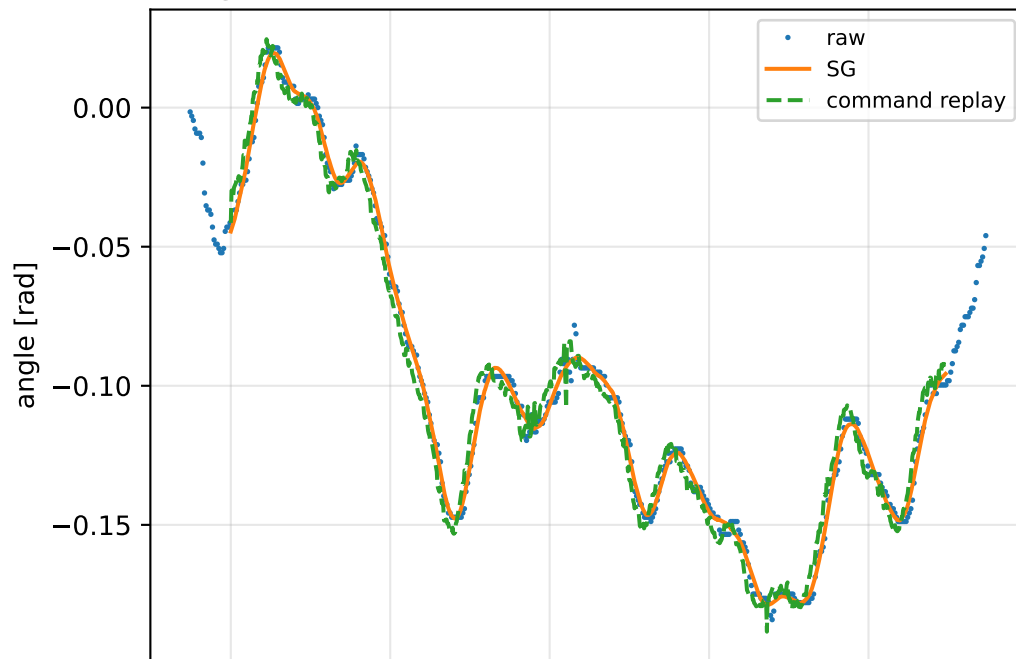


angular acceleration [rad/s²]

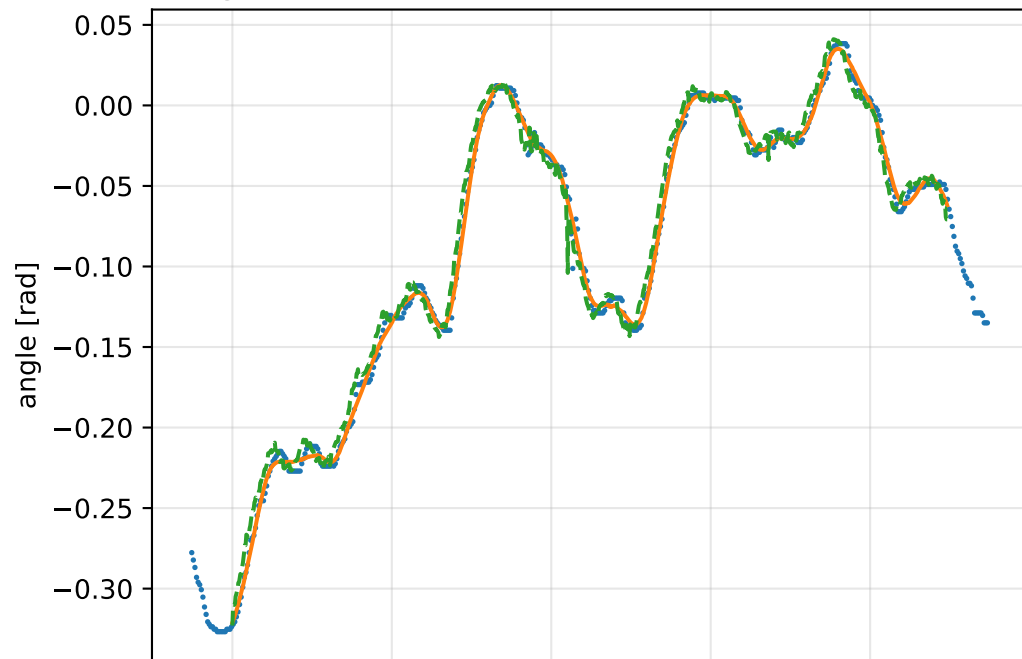


cog_all_nominal: gimbal measurement audit (objective=measured_sg)

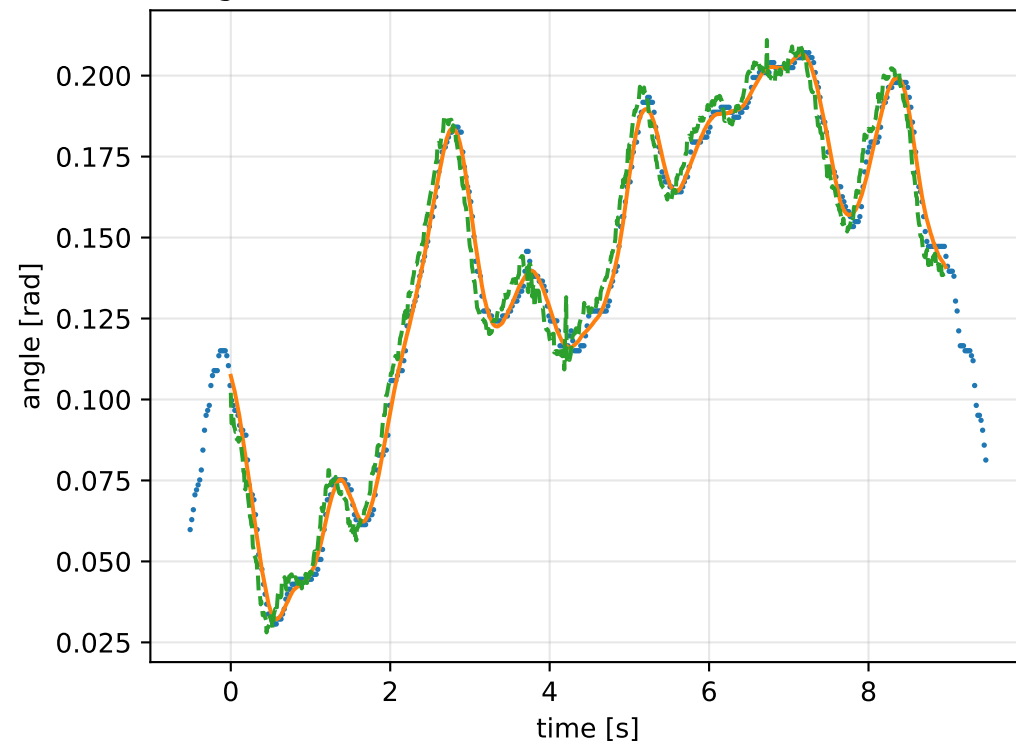
gimbal 1: RMSE=0.006345, max=0.01494 rad



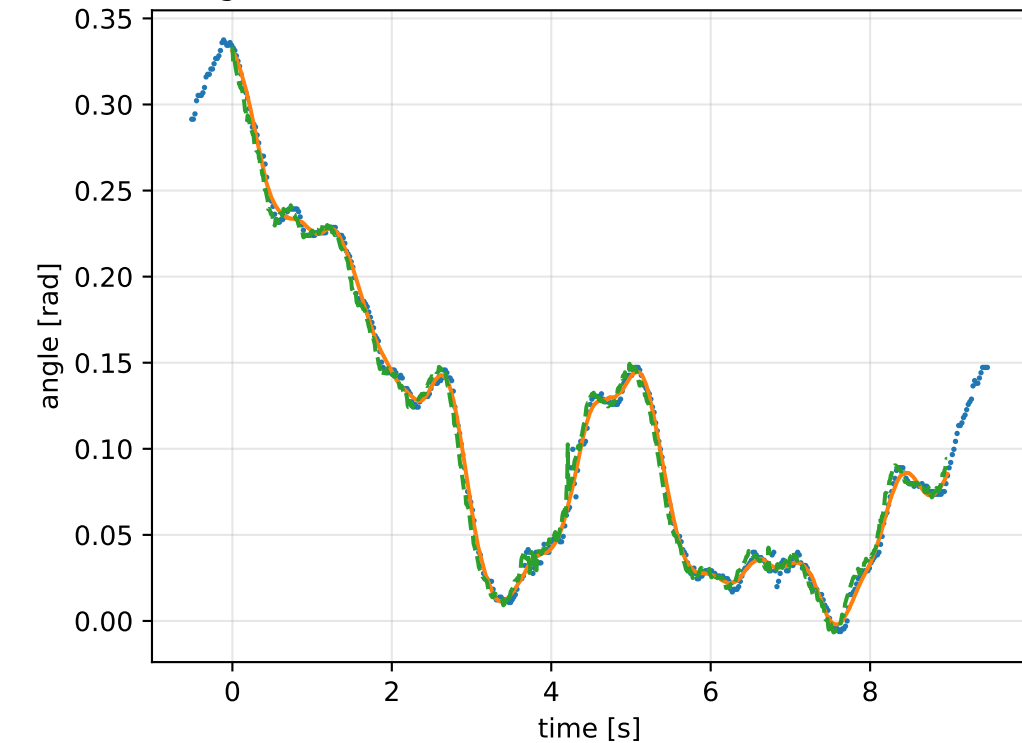
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

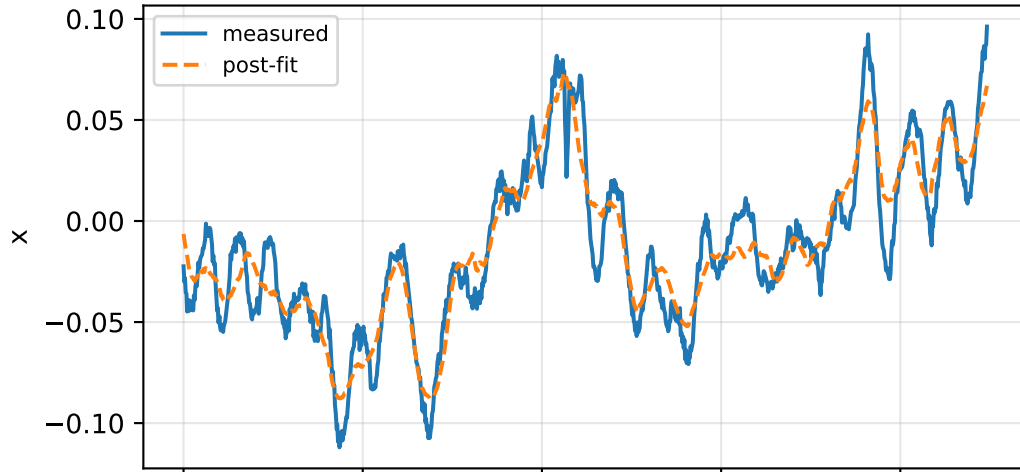


gimbal 4: RMSE=0.006321, max=0.03642 rad

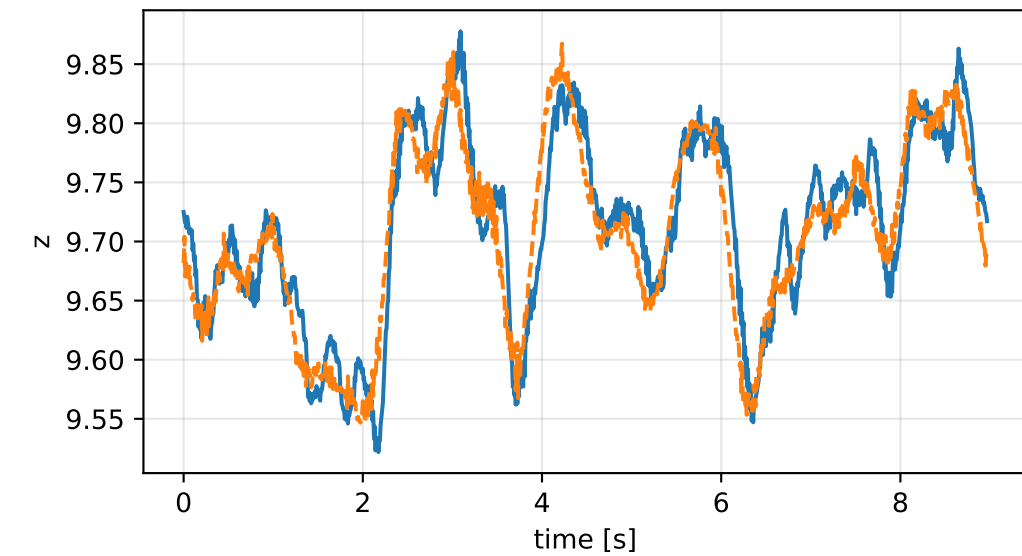
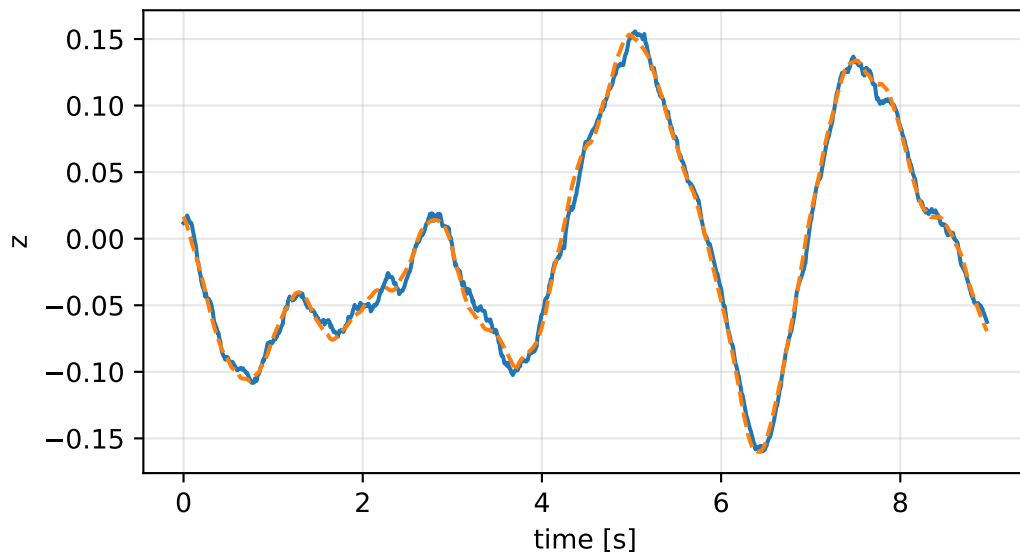
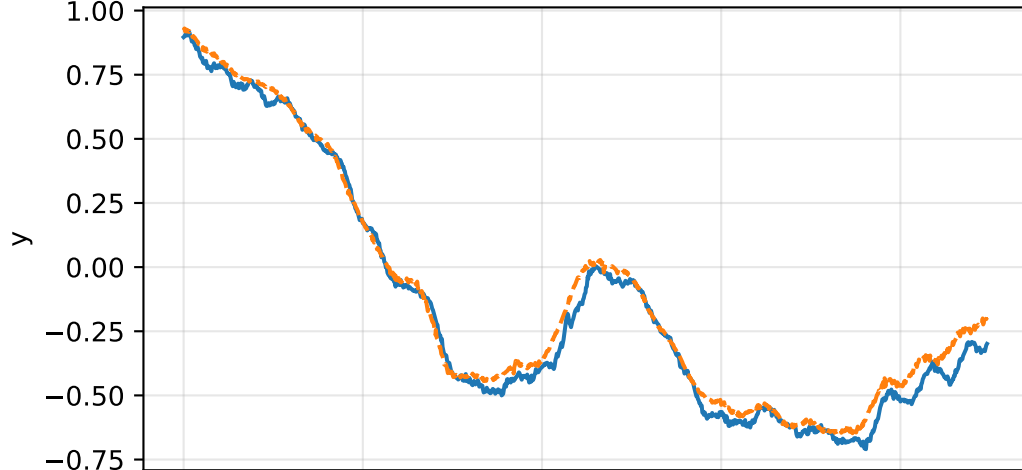
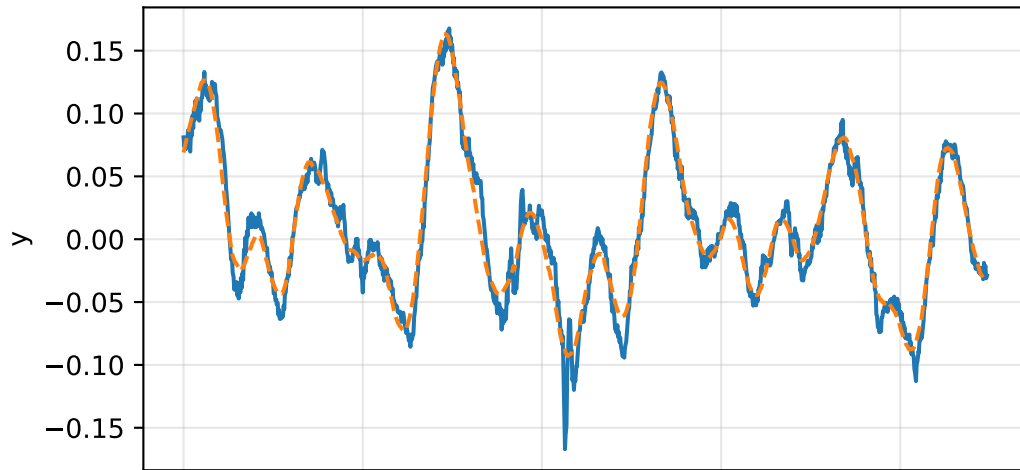
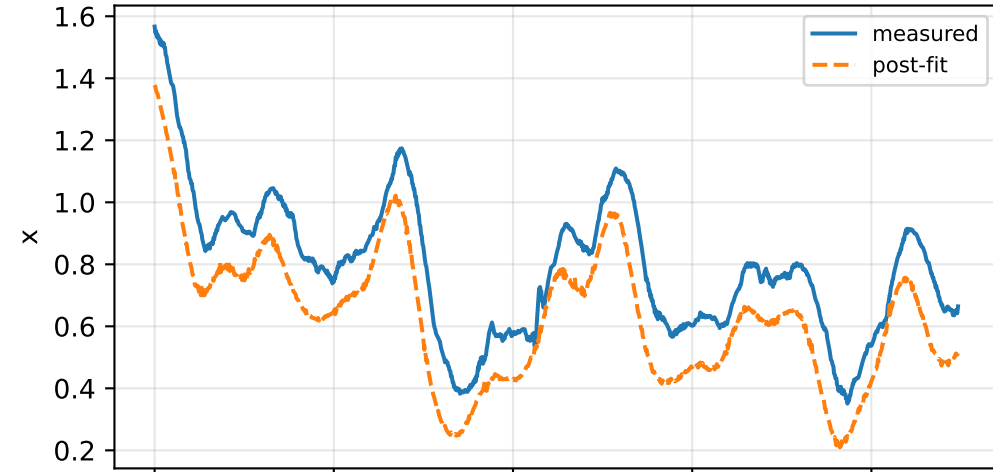


cog_all_nominal: IMU comparison (diagnostic only)

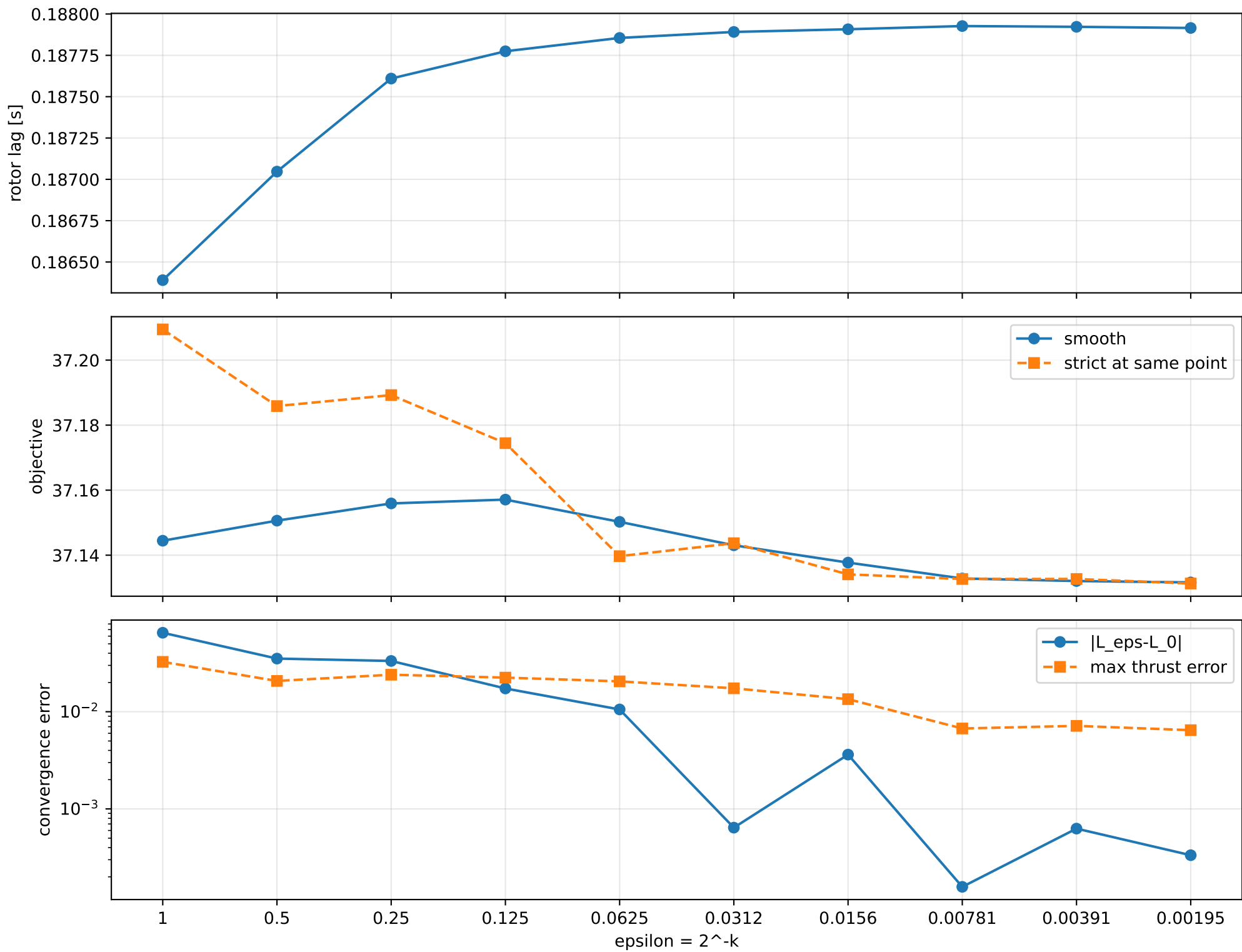
gyro [rad/s]



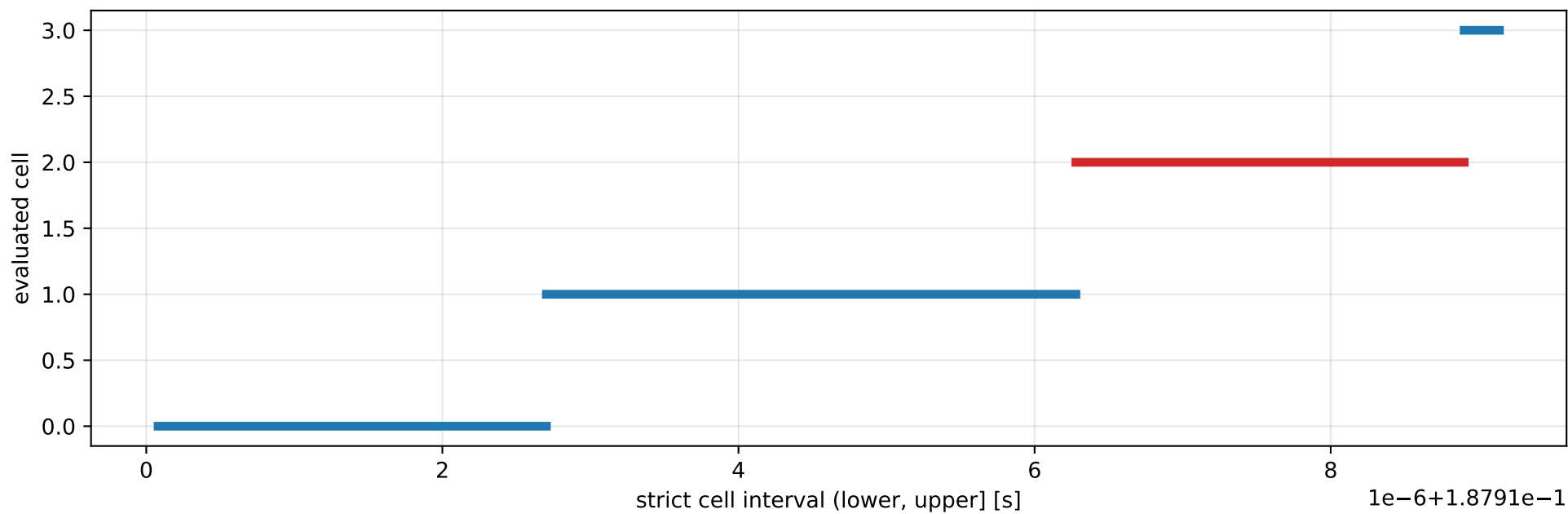
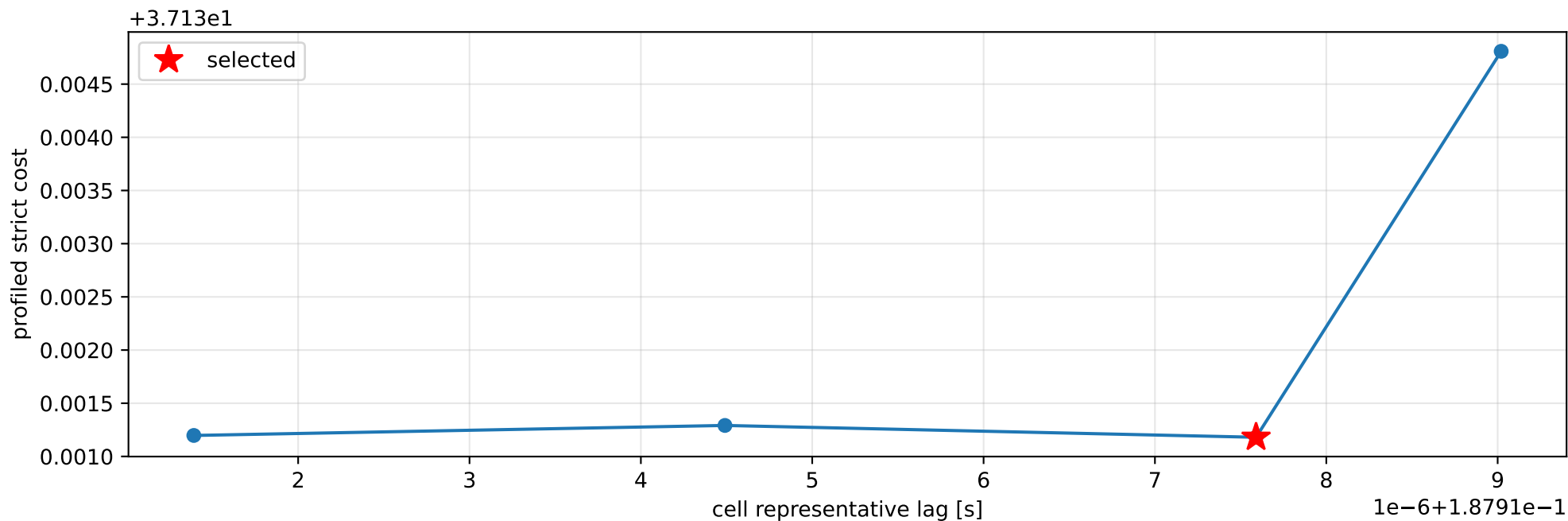
specific force [m/s^2]



cog_all_nominal: smooth-to-strict continuation



cog_all_nominal: exact strict-ZOH lag cells



selected (0.187916279, 0.187918901] s; final smooth 0.187915582 s; data boundary=False

cog_all_nominal: parameter estimate

Case: cog_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.96157266

inertia [kg m²]:

[[2.26600669 0.09096631 0.14063254]

[0.09096631 0.47991733 -0.70349894]

[0.14063254 -0.70349894 1.99778708]]

CoG body [m]: [-2.02470642e-03 -3.05351969e-05 9.50973625e-03]

force effectiveness: [0.65503406 0.74433465 0.6486374 0.6699967]

scale-free J/m [m²]:

[[1.15519896 0.04637417 0.07169377]

[0.04637417 0.24465947 -0.35864027]

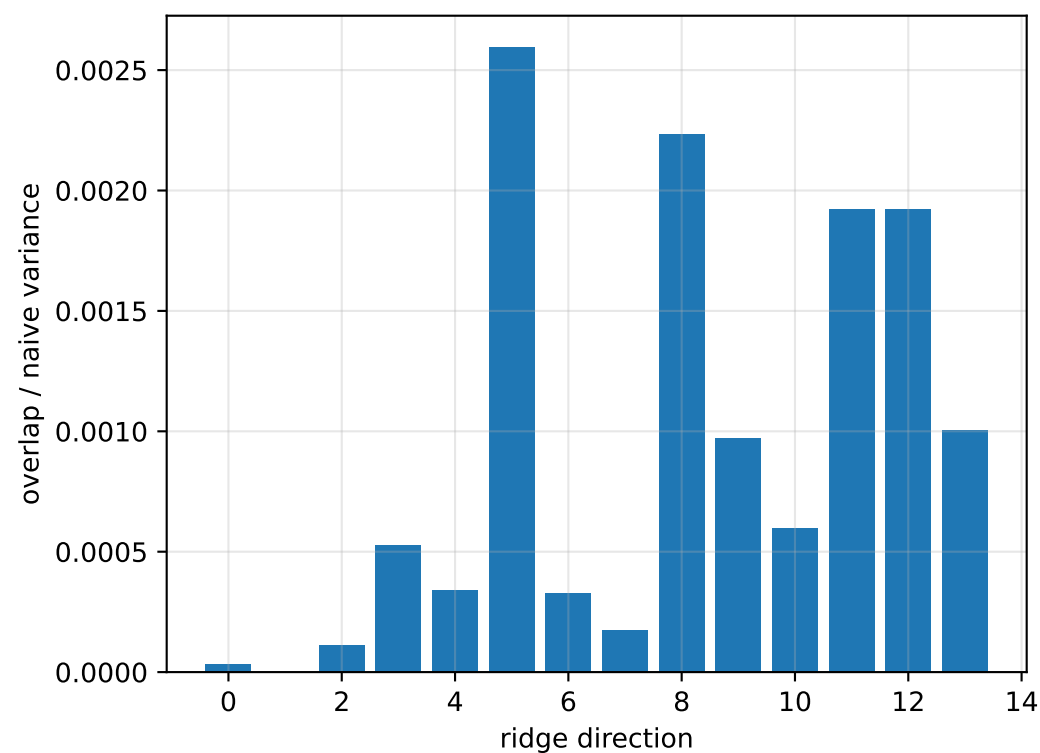
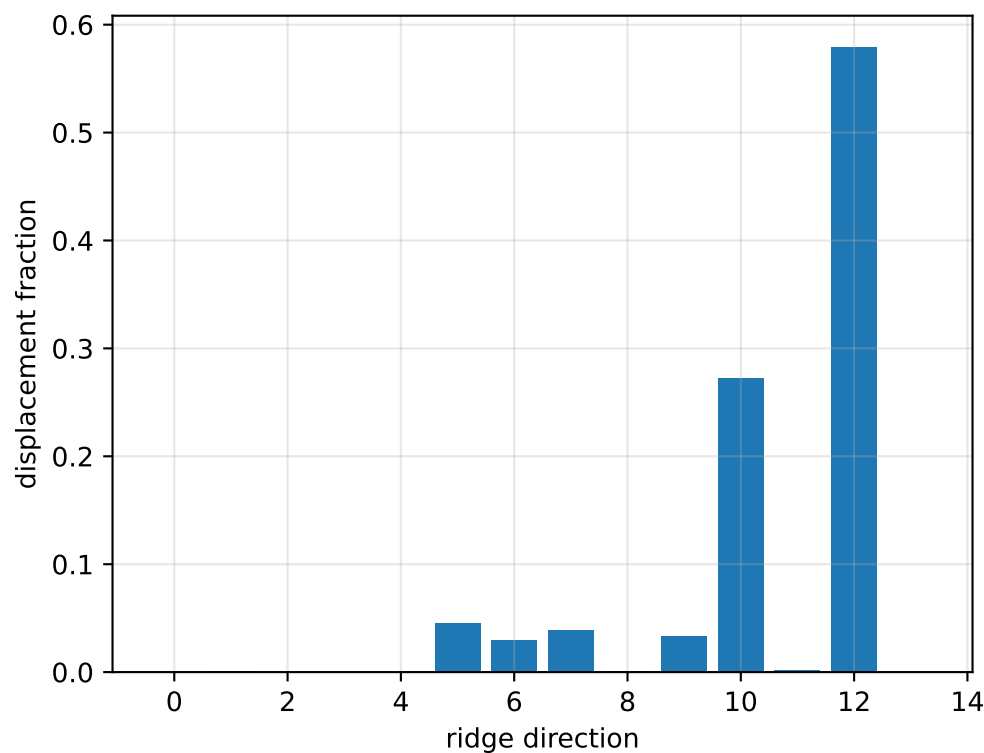
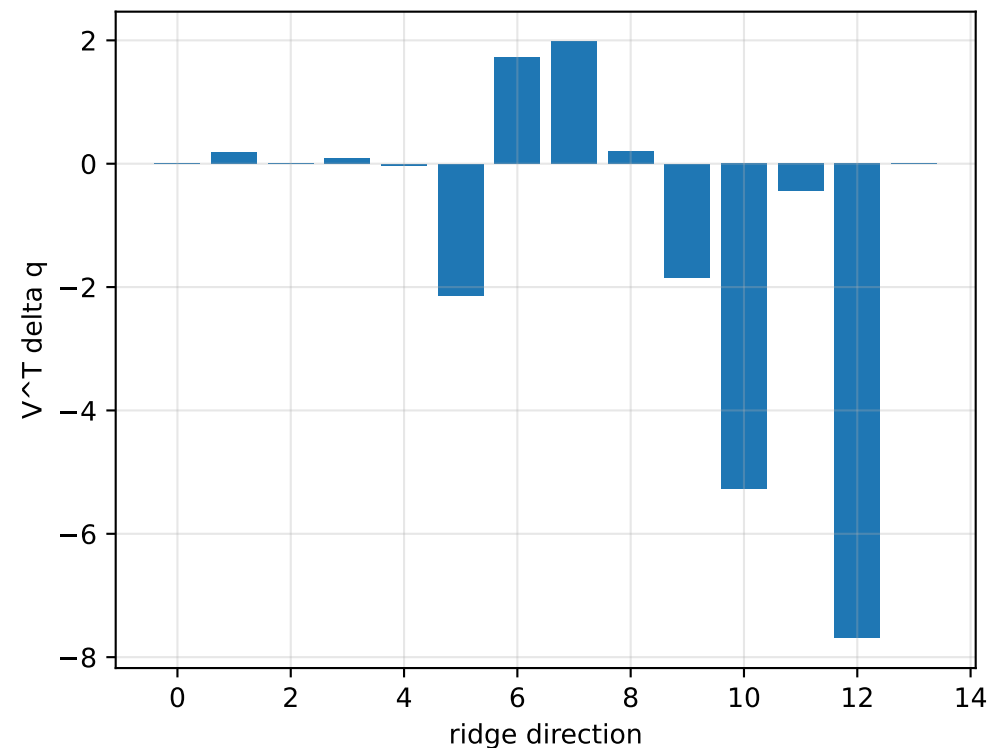
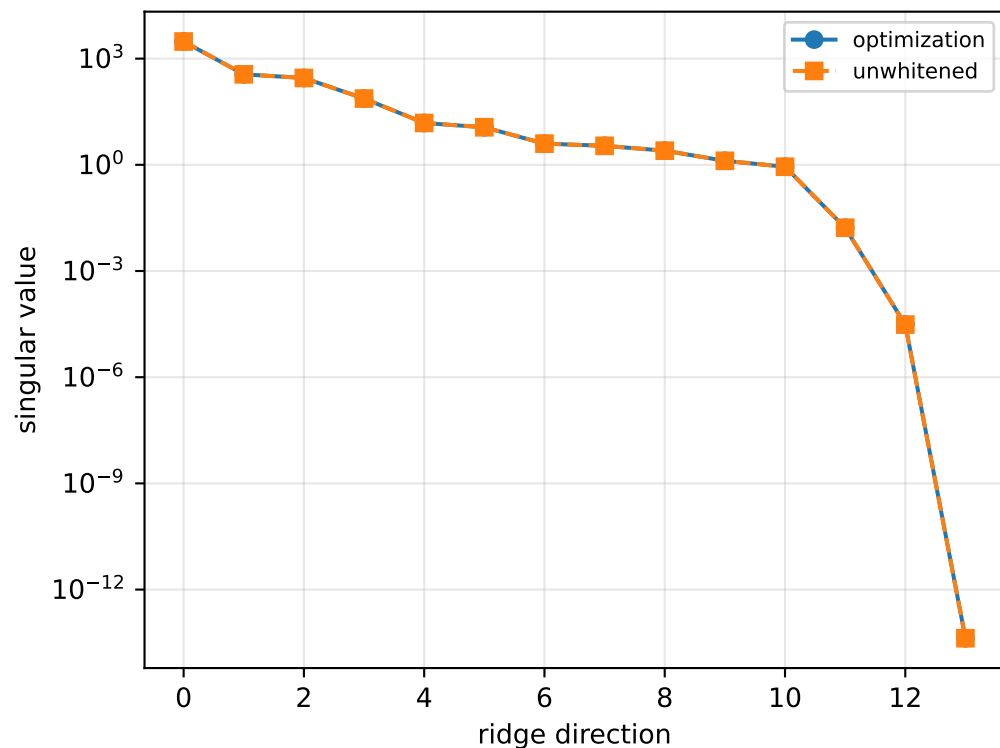
[0.07169377 -0.35864027 1.01846193]]

scale-free f/m: [0.33393311 0.37945811 0.33067213 0.34156099]

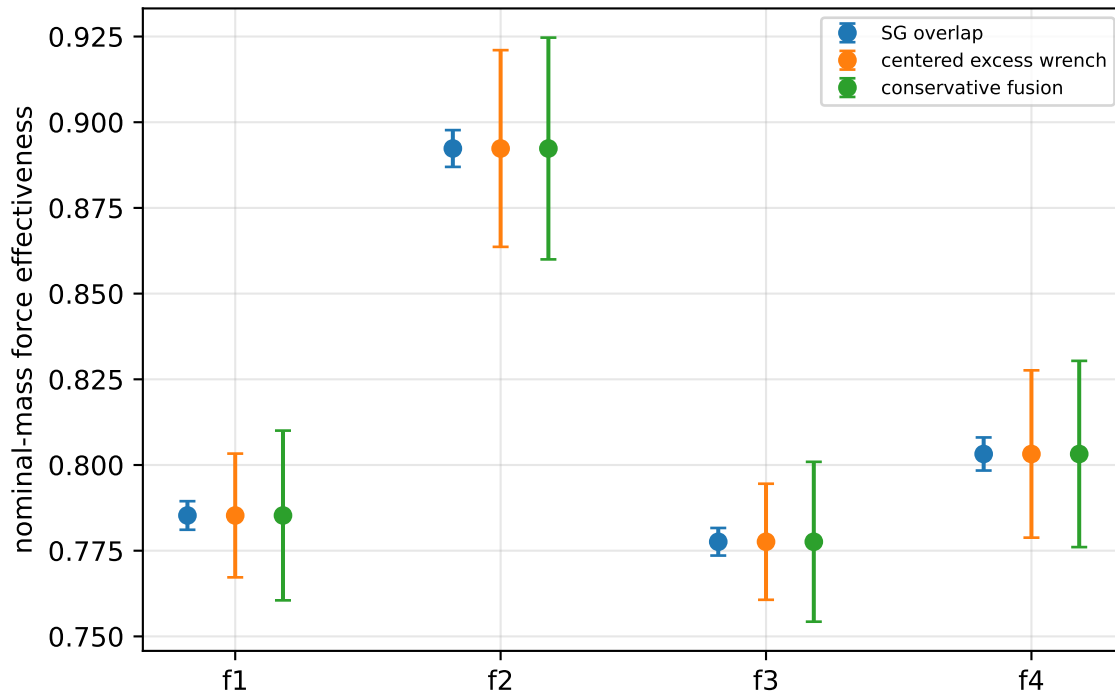
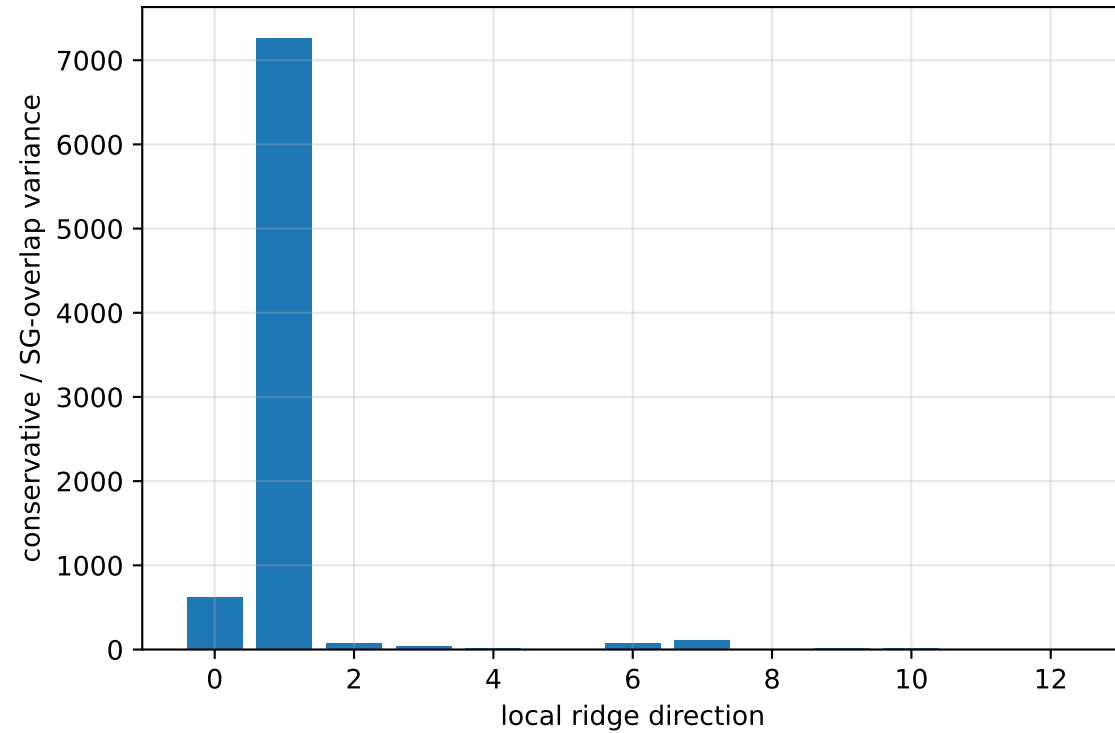
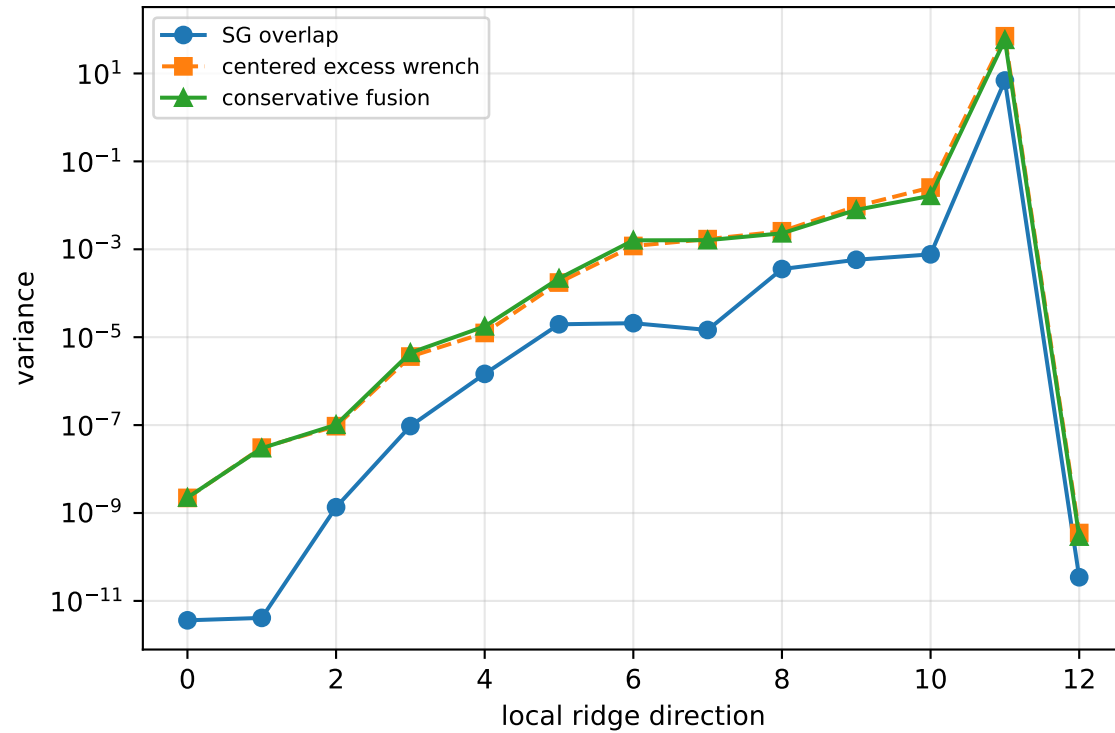
rotor lag cell: (0.187916279, 0.187918901] s

representative: 0.18791759 s

cog_all_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=8.2e-14)



cog_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 599.9$
wrench-to-acceleration recovery max error = $2.97\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

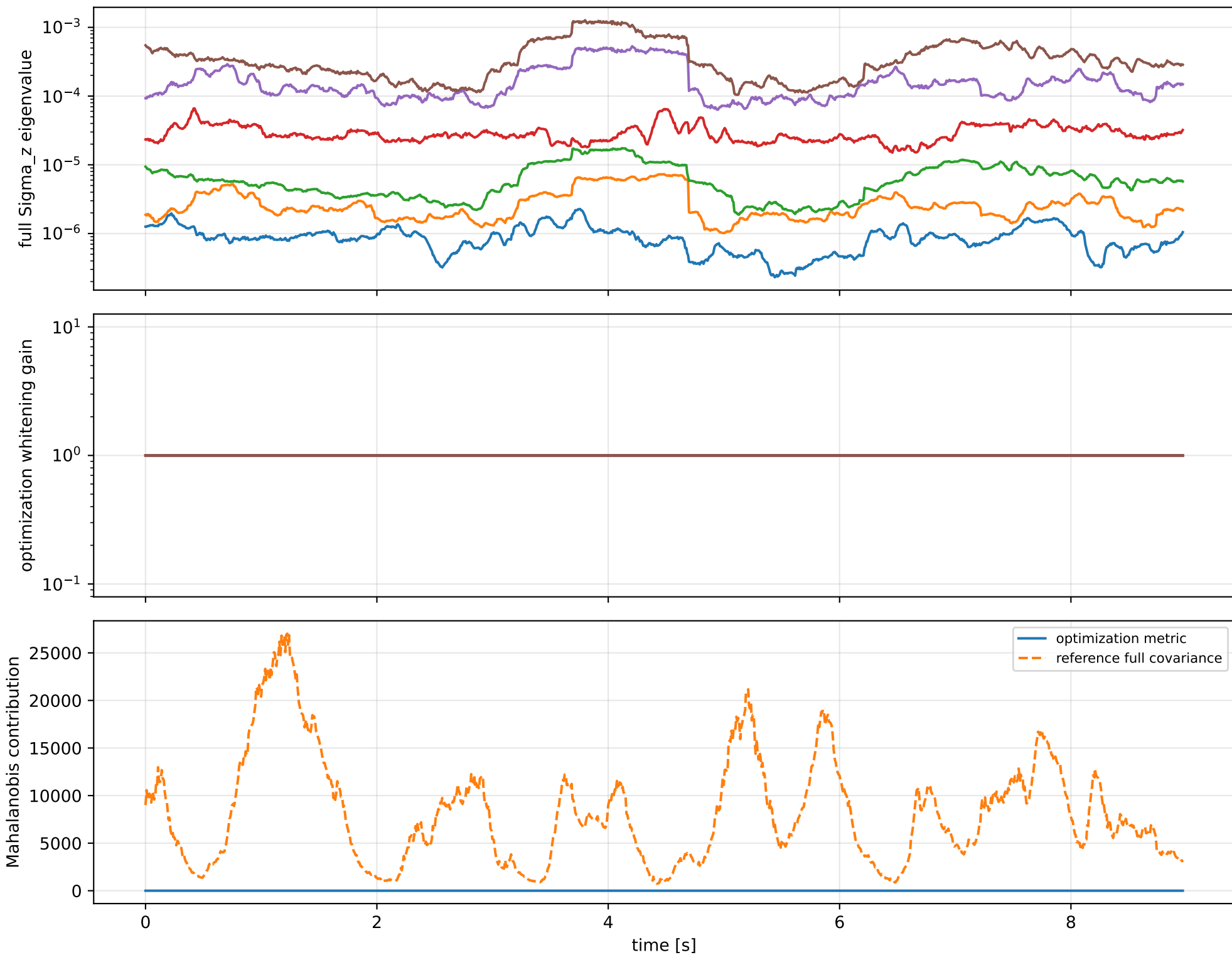
index 11: variance=58.18, ratio=8.41
 $\log_{\text{mass}}=+0.0249$; second_moment_diag_1=+0.394
second_moment_diag_2=-0.457; second_moment_diag_3=-0.0616
second_moment_offdiag_12=+0.751; second_moment_offdiag_13=-0.165
second_moment_offdiag_23=+0.196; cog_x=-0.000112
cog_y=+8.2e-05; cog_z=-8.98e-05
log_force_eff_1=+0.0249; log_force_eff_2=+0.024
log_force_eff_3=+0.0249; log_force_eff_4=+0.0258

index 10: variance=0.01644, ratio=21.45
 $\log_{\text{mass}}=+0.138$; second_moment_diag_1=+0.00368
second_moment_diag_2=+0.12; second_moment_diag_3=-0.808
second_moment_offdiag_12=-0.0744; second_moment_offdiag_13=-0.395
second_moment_offdiag_23=-0.115; cog_x=+0.0014
cog_y=-0.0289; cog_z=+0.0115
log_force_eff_1=+0.262; log_force_eff_2=+0.262
log_force_eff_3=+0.0171; log_force_eff_4=+0.0059

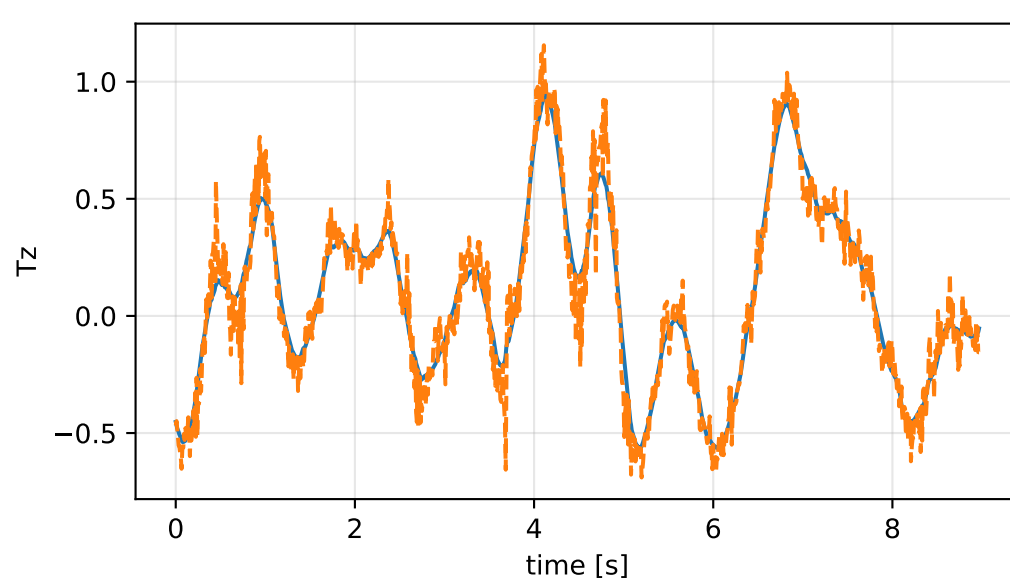
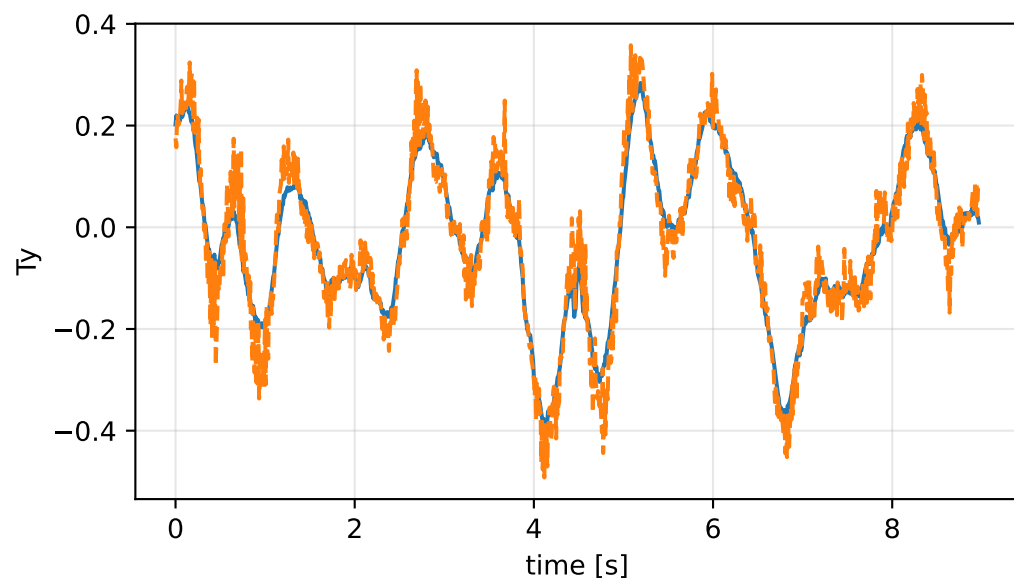
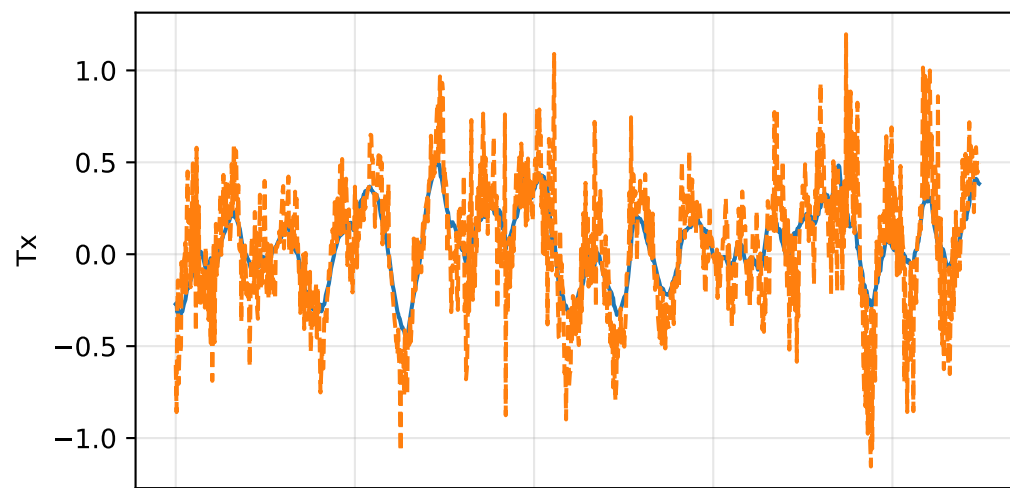
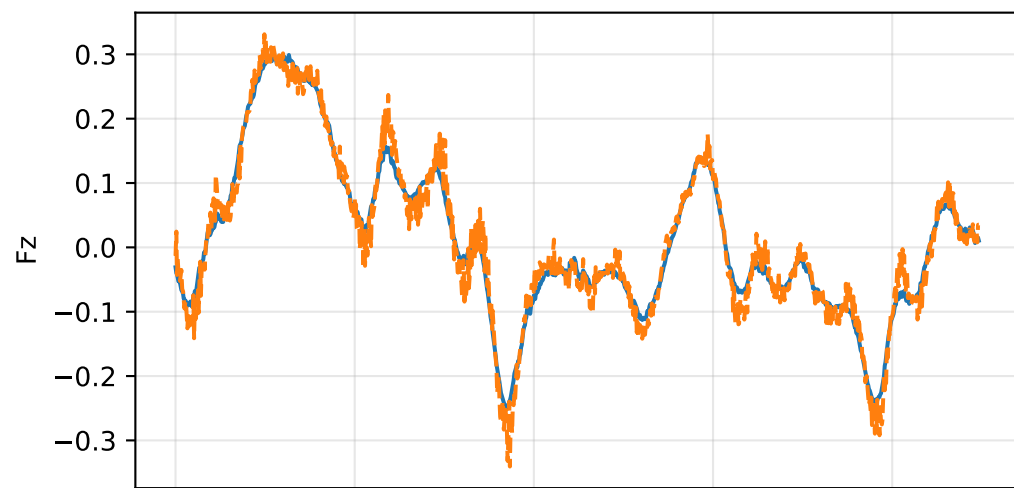
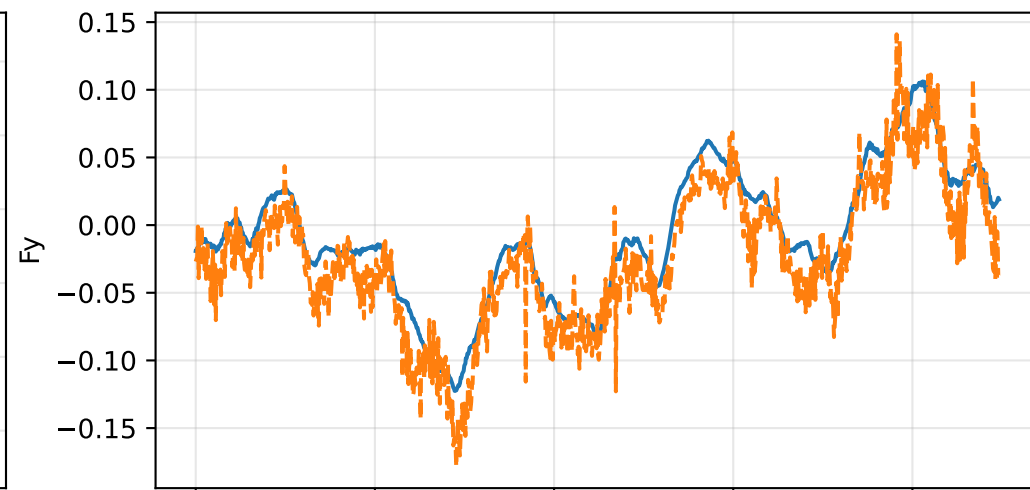
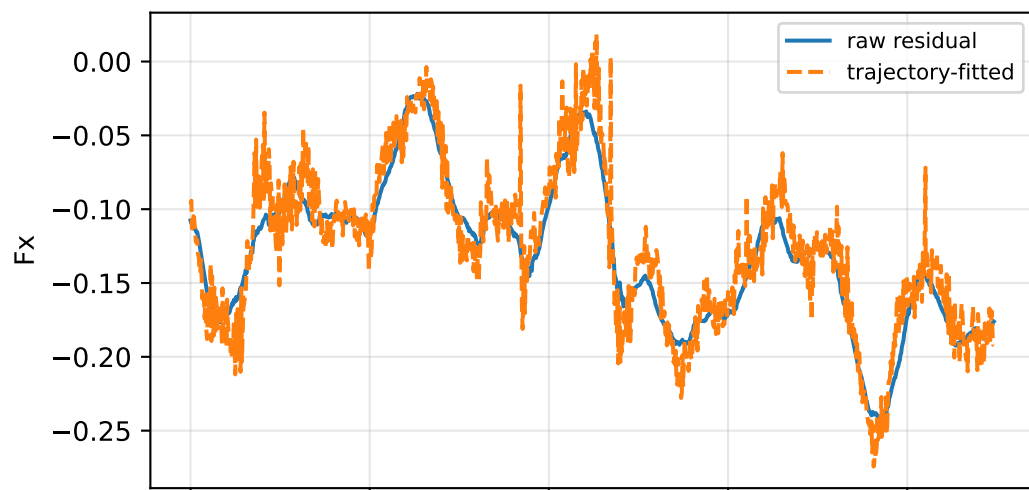
index 9: variance=0.007882, ratio=13.66
 $\log_{\text{mass}}=+0.0689$; second_moment_diag_1=-0.0137
second_moment_diag_2=-0.28; second_moment_diag_3=-0.0495
second_moment_offdiag_12=+0.109; second_moment_offdiag_13=+0.562
second_moment_offdiag_23=-0.626; cog_x=+0.0113
cog_y=-0.0474; cog_z=+0.0153
log_force_eff_1=+0.238; log_force_eff_2=+0.309
log_force_eff_3=-0.0912; log_force_eff_4=-0.181

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_all_nominal: optimization vs reference covariance



cog_all_nominal: wrench / closure (force max 5.83e-15, torque max 2.22e-16)



cog_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_cog_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/cog_all_nominal.

SHA256: c8e09152b70075c9dea19c9858059b27978f15fa2ccc061d3e475ff0b4b7566a

data objective: 37.1311788605

prior objective: 1.2848150877e-06

total objective: 37.1311801453

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog_all_nominal (cog_position_body_m)

components: x, y, z

target: [-2.02470856e-03 -3.05265789e-05 9.50974960e-03]

std: [1.e-05 1.e-05 1.e-05]

final: [-2.02470642e-03 -3.05351969e-05 9.50973625e-03]

error: [2.14413925e-09 -8.61794946e-09 -1.33452850e-08]

standardized residual: [0.00021441 -0.00086179 -0.00133453]

factor objective: 1.2848150877e-06